Baseline Sleep Characteristics in NCAA Division I Collegiate Athletes

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Abstract

Objective: The authors report no conflicts of interest.

To determine baseline sleep characteristics of male/female student-athletes across multiple sports using objective and subjective measures. **Design:** Prospective study. **Setting:** Division I college. **Participants:** Eighty-two male and female Division I student-athletes. **Interventions:** Participants completed 2 validated sleep questionnaires (Epworth Sleepiness Scale [ESS] and Single-Item Sleep Quality Scale [SISQS]) to assess subjective sleep. They also wore a validated sleep monitoring device (WHOOP 4.0 band) for at least 14 nights to collect objective data on total sleep time (TST) and sleep architecture. **Main Outcome Measures:** Overnight sleep variables, including TST, time spent awake in bed after falling asleep, time spent in light sleep, rapid eye movement (REM) sleep, and slow-wave sleep (SWS) cycles. Sleep quality and daytime sleepiness were also assessed. **Results:** There were no statistical differences between male and female student-athletes in average TST, sleep architecture, sleep consistency, SISQS, and ESS scores. The average TST was 409.2 ± 36.3 minutes. Sleep architecture consisted of 25.6% REM, 19.9% SWS, and 54.4% light sleep. The average sleep consistency was 61.6% ± 8.9%. The average SISQS score was 6.48 ± 1.71, and the average ESS score was 7.57 ± 3.82. A significant difference was found in average wake time between males and females (55.0 vs 43.7 min, $P = 0.020$), with an overall average of 50.2 ± 16.2 minutes. **Conclusions:** College student-athletes do not typically obtain the recommended amount of sleep. Optimizing sleep can positively affect academic and athletic performance.

Key Words: sleep, quality, restriction, sports, student-athlete

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INTRODUCTION

Sleep is critical for the health and well-being of individuals. The American Academy of Sleep Medicine recommends adults receive 7 to 9 hours of sleep per night and adolescents obtain 8 to 10 hours per night for optimal health.[1] In more physically active populations, such as athletes, it has been suggested 9 to 10 hours of sleep is required to allow for adequate recovery and peak performance.[2] Disturbed sleep states, such as sleep deprivation or restriction, have been shown to negatively affect mood, daytime sleepiness, athletic performance, immune response, and recovery in athletes.[3–8] This has led to increased attention being placed on sleep in athletes, with the National Collegiate Athletic Association (NCAA) Interassociation Task Force and the International Olympic Committee recognizing sleep issues and disorders as important factors affecting mental and physical health in athletes.[9,10]

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Within the athletic population, college athletes represent a unique demographic where sleep is affected by a combination of academic and athletic demands. In 2018, Mah et al[11] surveyed 628 collegiate athletes across 29 different sports and found that 39.1% had inadequate sleep, averaging less than 7 hours of sleep on weekdays, with 42.4% reporting poor sleep quality and 51% reporting high levels of daytime sleepiness. This is similar to the findings of Rabin et al[12] who subjectively evaluated sleep using a self-reported validated questionnaire in 1055 collegiate athletes across 4 institutions and found over 65% had suboptimal sleep health, and 23.7% had clinically meaningful problems with their sleep. These studies suggest collegiate athletes have decreased sleep time and sleep quality through subjective measures, but few studies have evaluated sleep metrics using subjective and objective measures in this population.

To our knowledge, there have only been 2 studies to date that have simultaneously captured subjective and objective sleep data in collegiate athletes. In 2011, Mah et al[13] found that 11 collegiate basketball players had decreased sleep time through actigraphy (the recording of the activity level of the body using a small device typically worn on the wrist that is used to measure the amount and quality of sleep) and a sleep diary that was consistent with insufficient sleep in the general public. More recently, Carter et al[14] evaluated sleep using actigraphy and a sleep diary in 121 Division II collegiate athletes from 6 team sports and 3 individual sports and found via actigraphy that 94% of athletes received <8 hours of sleep per night, with 61% having <7 hours. Nearly 44% of athletes

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in this cohort also subjectively reported symptoms of mild insomnia and/or increased daytime sleepiness. Although the findings of Mah and Carter demonstrate decreased sleep in collegiate athletes through objective metrics, there still remains a general paucity of sleep data on collegiate athletes across multiple sports at the Division I level.

Therefore, the primary aim of this study was to determine baseline sleep characteristics of male and female NCAA Division I collegiate athletes across multiple sports using objective measures (sleep duration, sleep architecture) and subjective measures (sleep quality, daytime sleepiness). It was hypothesized that male and female athletes in this population would demonstrate insufficient sleep patterns with decreased total sleep time (TST), poorer sleep architecture, decreased sleep quality, and increased daytime sleepiness.

METHODS

Subjects

Eighty-two varsity athletes from University of California, Los Angeles, volunteered to participate in this study, including 47 male and 35 female athletes. Participants were tested at various times throughout the year, including in-season and during the "off-season," which varied in time of year, practice hours, and/or academic course work. Our sample was comprised of student-athletes from the following collegiate sports: women's basketball (n = 7), men's basketball (n = 3), women's softball (n = 4), men's baseball (n = 18), women's soccer (n = 16), men's soccer (n = 4), men's tennis (n = 1), women's swimming (n = 8), and men's water polo (n = 21). This study was approved by the University of California, Los Angeles, institutional review board, and all participants provided written informed consent before participation.

Experimental Design

Subjects completed 2 validated questionnaires to evaluate subjective sleep, including the 8-question Epworth Sleepiness Scale (ESS), and the Single-Item Sleep Quality Scale (SISQS).[15,16] These questionnaires have been validated in the general population, but not in a collegiate athlete population.

After the completion of all questionnaires, participants wore a validated sleep monitoring device (WHOOP 4.0 band, *WHOOP*, Boston, MA) that collected TST, sleep architecture, and sleep consistency.

Measurements

Wearable Sleep Device

The WHOOP 4.0 band is a wearable fitness and sleep tracker that uses accelerometers and photoplethysmography to provide 2-stage and 4-stage categorization of sleep. It has been validated against polysomnography and actigraphy for quantification of sleep in healthy volunteers.[17–19] Objective overnight sleep variables included time spent in light sleep, rapid eye movement (REM) sleep, and slow-wave sleep (SWS) cycles, time spent awake in bed after falling asleep (wake time), and TST defined as the sum of light sleep, REM, and SWS durations. Study inclusion criteria required a *minimum* of 14 days of recorded sleep data and, on average, participants provided 67.7 days of recorded sleep data (67.7 ± 62.1 days).

Sleep Consistency

This study defines sleep consistency using a proprietary metric of the WHOOP platform adapted from the Sleep Regularity Index (SRI) for daily use by accounting for recency in weighting of comparator sleep-wake episodes.[20,21] The WHOOP sleep consistency measure, such as the SRI developed by Phillips et al, calculates the percentage of concordance of individuals being in the same state (asleep vs awake) at different time points. Whereas the SRI compares 2 timepoints 24 hours apart, WHOOP sleep consistency compares 24 hours time points over a 4-day interval (eg, timepoint 1 [T1], T1 + 24 hours, T1 + 48 hours, T1 + 72 hours), with comparisons of intervals further apart assigned progressively lower weights in calculating sleep consistency scores for a given time point.[22] Scores are converted to a 0%-100% scale, with higher consistency scores reflecting lower sleep timing variability.

Data Analysis

The data were aggregated via means per person, so each participant was only considered once. Subjective and objective sleep data were checked for normality using the Shapiro–Wilk test. If data had a normal distribution, a *t* test was used to compare between genders; if data had a non-normal distribution, a Wilcoxon Rank Sum test was used to compare between genders. Data are presented as mean ± SE for data that met assumptions of normality. The significance level was set at $\alpha = 0.05$.

Ethical Considerations

With all research involving biometric data, there must be significant safeguards in place to protect patient confidentiality. Specifically in an athletic population, coaching staff are increasingly interested in monitoring athlete biometrics including sleep data. We must keep in mind that, although wearable technology such as those used in this study provide in-depth insights into athlete health and recovery, these data are protected health information and must be treated as such. Caution must be taken in implementing biometric monitoring programs in the athletic setting, which involve informed consent, institutional elimination of coercion, and subsequent safeguarding of protected patient information.

RESULTS

Objective Sleep Data

There were a total of 82 undergraduate student-athletes from 2021 to 2022 who satisfied the inclusion criteria of 14 consecutive nights of recorded sleep. The demographics were 57% male (n = 47) and 43% female (n = 35). Comprehensive sleep characteristics of the subjects are presented in Figure 1. The average TST was 409.2 ± 36.3 minutes, with no statistically significant difference between male and female subjects (403.8 vs 416.5 min, $P = 0.215$). The average wake time was 50.2 ± 16.2 minutes, with a statistically significant difference noted between male and female subjects (55.0 vs 43.7 min, $P = 0.020$).

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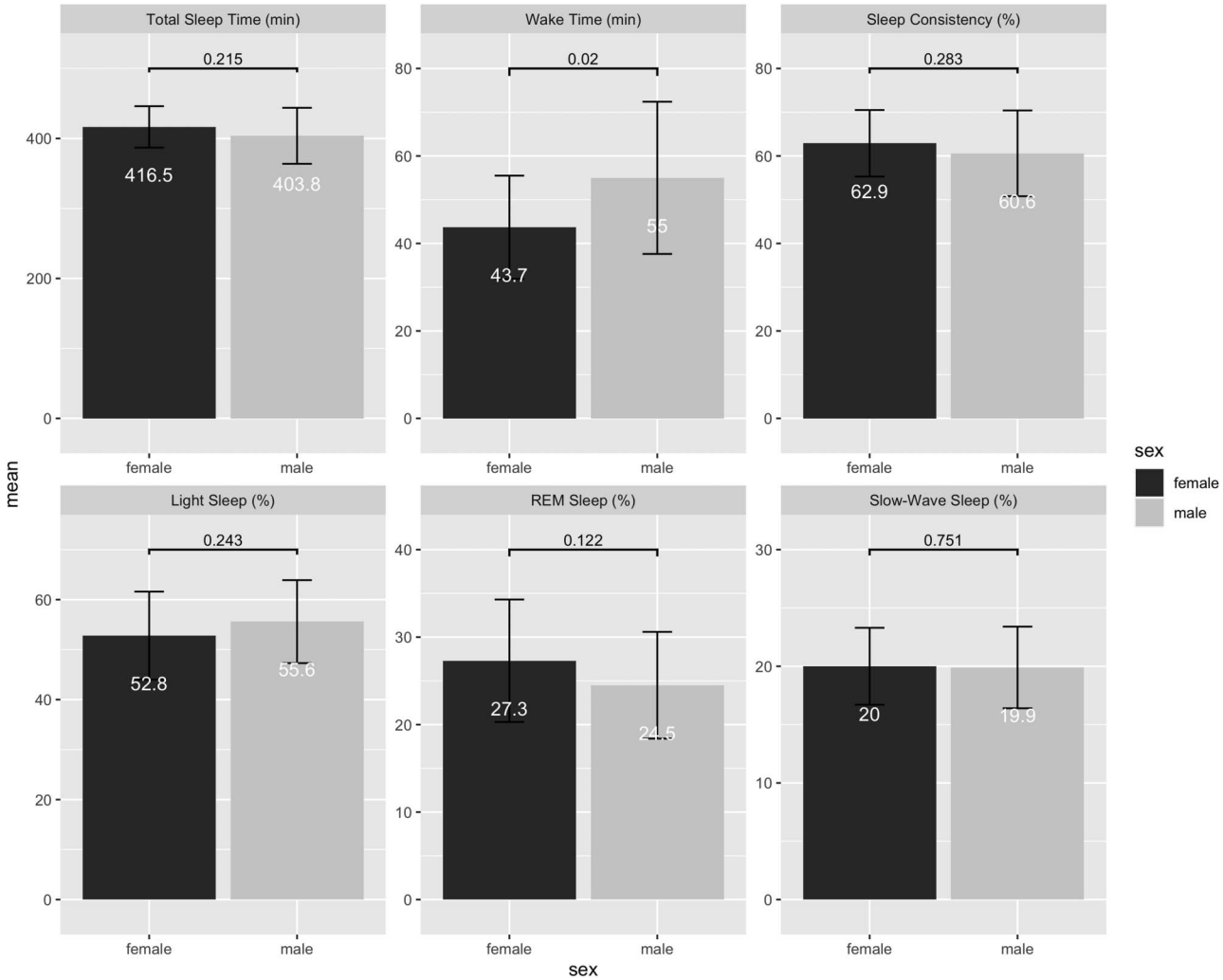

Figure 1. Sleep characteristics. Average times for each sleep phase (light sleep, REM sleep, slow-wave sleep), wake time, and TST are reported in minutes and compared by sex. Average sleep times are presented within the appropriate bar graph. *P*-values for sex differences within each subcategory of sleep are presented above the comparator bars. Sleep consistency, calculated as the percentage of concordance of individuals being in the same state (asleep vs awake) at different timepoints, is compared by sex and reported as a percentage.

The average sleep architecture distribution was 25.6% REM, 19.9% SWS, and 54.4% light sleep, with no statistically significant difference between male and female subjects (24.5% vs 27.3%, $P = 0.122$; 19.9% vs 20.0%, $P = 0.751$; 55.6% vs 52.8%, $P = 0.243$).

The average sleep consistency was 61.6% ± 8.9%, with no statistically significant difference between male and female subjects (60.6% vs 62.9%, $P = 0.283$).

Subjective Sleep Survey

There were a total of 82 respondents (47 male, 35 female) for the SISQS and the ESS. The average SISQS score was 6.48 ± 1.71, with no statistically significant difference between male and female subjects (6.42 vs 6.61, $P = 0.643$). The distribution of SISQS scores in the cohort is presented in Figure 2. An SISQS score of 6.48 is interpreted as "fair to good sleep quality" and the SISQS scoring system is presented in Table 1. The average ESS score was 7.57 ± 3.82, with no statistically significant difference between male and female subjects (7.15 vs 8.65, $P = 0.111$). The distribution of ESS scores in the cohort is presented in Figure 3. An ESS score of 7.57 is interpreted as "higher normal daytime sleepiness" and the ESS scoring system is presented in Table 2.

DISCUSSION

This study aims to quantify the baseline sleep habits of NCAA Division 1 collegiate athletes, which will serve as a foundation on which clinicians can build informed sleep intervention recommendations. The use of novel wearable technology to quantify sleep adds validity and objectivity to the data, which has not been conducted in a cohort of this size (n = 82) for an extended time period (67.7 days of recorded sleep data on average).

Studies have shown that college students and athletes often have poor baseline sleep patterns.[23–25] A study of 1125 college students aged 17 to 24 years found that 60% of students were considered "poor-quality sleepers" based on the PSQI.[23] They found that their bedtimes and rise times were delayed during weekends and that students frequently took prescription, over-the-counter, and recreational psychoactive drugs to help

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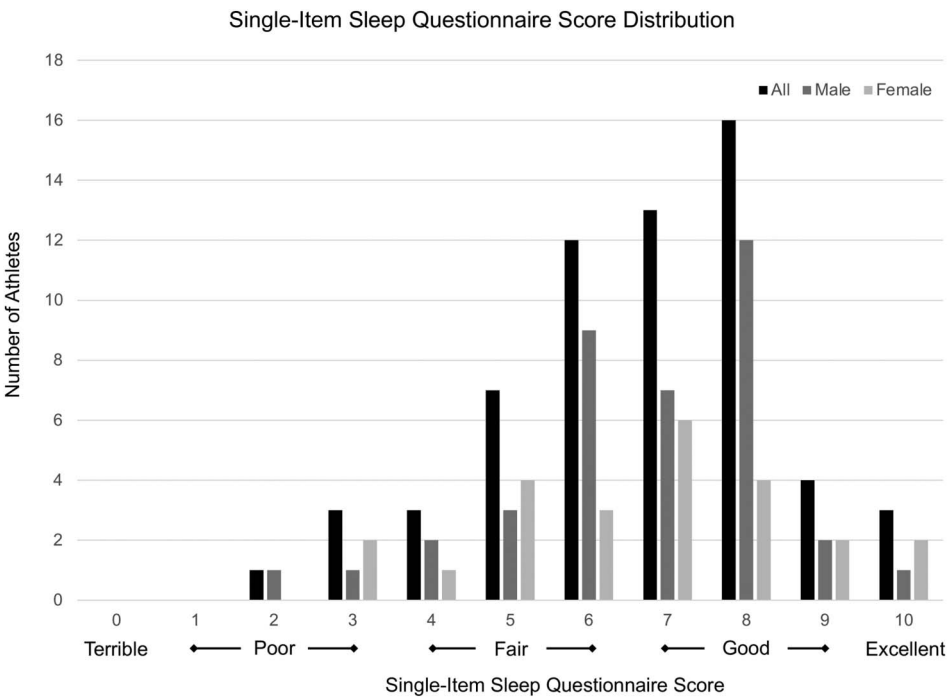

Figure 2. SISQS Score distribution. The number of athletes reporting each value on the SISQS are reported in aggregate (black bars) and concurrently by sex (male = dark gray bars, female = light gray bars).

with sleep. A study published in the Journal of Athletic Training found that student-athletes reported an average of 6.9 hours of sleep per night, with 30% reporting less than 6 hours of sleep per night.[24] Another study published in the Journal of Sports Sciences found that elite athletes reported an average of 7.1 hours of sleep per night, with 25% reporting less than 6 hours of sleep per night.[26] Moreover, Hicks et al[27] showed a marked increase in sleep dissatisfaction among college students over the past 20 years.

Our study found a TST of 6.8 hours per night in collegiate athletes, which parallels findings of previous studies. The use of a wearable device to quantify sleep rather than self-reported sleep diaries adds validity and objectivity to our findings. Research has repeatedly shown that self-reported sleep diaries are inaccurate. A study in the Journal of Sleep Research found that sleep duration measured with the Pittsburgh Sleep Quality Index or sleep diary was about 2 hours above or up to 4 hours below polysomnography-measured sleep duration.[28] In addition, wearable devices have shown accuracy similar to polysomnography. Lehrer et al[29] found that actigraphy and polysomnography produced similar estimates of sleep duration and efficiency, whereas sleep diaries yielded higher estimates of TST.

TABLE 1. SISQS Scoring System

SISQ Score	Score Interpretation
0	Terrible
1-3	Poor
4-6	Fair
7-9	Good
10	Excellent

Sleep consistency is a vital yet often overlooked component of sleep. Sleep consistency refers to the routine or regularity of sleep schedules and can be reported as intraindividual day-to-day variability in bedtimes/wake-up times or weekday-to-weekend variability in bedtimes/wake-up times.[30] Our study found an average sleep consistency of 61.6%, which is below average compared with a large cohort of adults tracked via the WHOOP device (average consistency of 67.9%).[22] Notably, sleep consistency seems to affect academic performance. A study of college students found a significant negative correlation between sleep inconsistency and overall academic performance, indicating that the greater inconsistency in sleep duration was associated with a lower GPA.[31] The literature has not shown, however, whether sleep consistency affects athletic performance.

Barriers to Sufficient Sleep

Simply extending the time that college student athletes sleep each night, although scientifically sound, may not be practical for a busy student-athlete. College athletes face a number of challenges when it comes to optimizing their sleep schedules, which can include[32]:

1. Balancing academic and athletic demands: College athletes are often required to maintain a certain grade point average to be eligible to participate in their sport. This requires that they attend classes, complete assignments, and study for examinations while also participating in practices, games, and other athletic events.
2. Travel: College athletes often travel for competition, which can disrupt their normal schedule and make it difficult to keep up with academics.
3. Training and conditioning: College athletes are required to spend a significant amount of time training and conditioning including strength and conditioning, practices, and recovery time for injuries.

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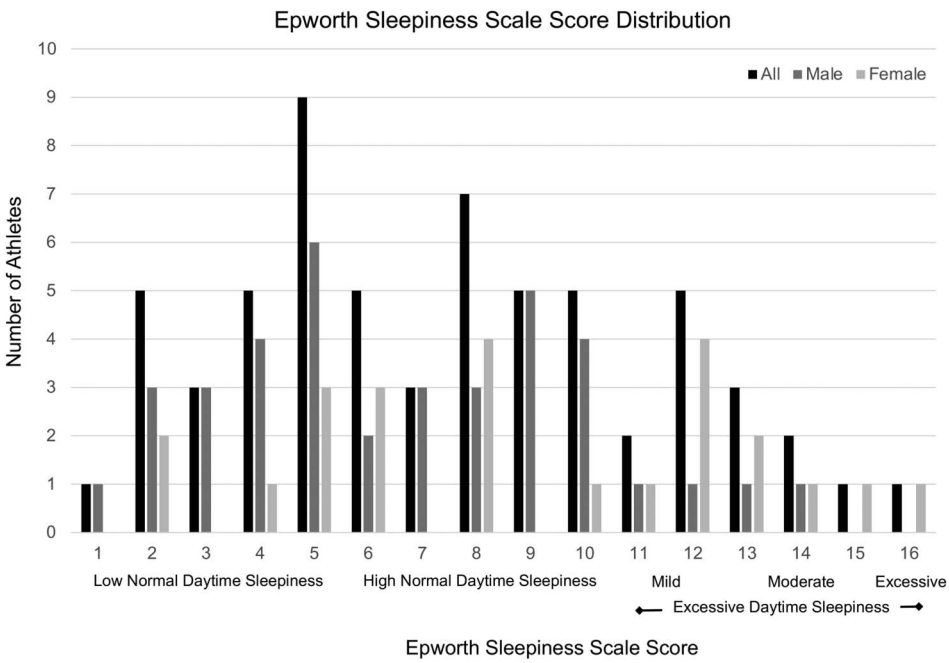

Figure 3. ESS Score distribution. The number of athletes with each ESS score are reported in aggregate (black bars) and concurrently by sex (male = dark gray bars, female = light gray bars).

4. Social life: College athletes also face challenges when it comes to maintaining a social life. They may not be able to participate in social events and activities as much as their nonathlete peers because of practice, games, and travel schedules.

The Impact of Reduced Sleep

Poor sleep habits can predispose college students to a number of medical issues including an increased risk of injury and illness.[33] A longitudinal study of collegiate volleyball players found that sleep duration was significantly worse the day before an injury occurred.[34] Other studies investigating sleep deprivation and injury risk have not found significant associations although they uniformly used self-reported TST, which has been shown to overestimate sleep duration.[35–38]

Illness seems to also be linked to sleep deprivation. One study of cellular immune function found a reduction of mature neutrophils in sleep-deprived male subjects.[39] Extrapolating these findings to the collegiate population, a study of 11 942 college students found that late bedtime was positively associated with self-reported short- and long-term illness.[40]

Moreover, sleep deprivation can have profound effects on cognition, which affects academic and athletic performance.

TABLE 2. ESS Scoring System

ESS Score	Score Interpretation
0-5	Lower normal daytime sleepiness
6-10	Higher normal daytime sleepiness
11-12	Mild excessive daytime sleepiness
13-15	Moderate excessive daytime sleepiness
16-24	Severe excessive daytime sleepiness

Data from the National College Health Assessment survey that included 112 849 college students found that insufficient sleep was associated with an 8% increased likelihood of a B average for each day of insufficient sleep and a 21% increased likelihood of a C and D/F average for each day of insufficient sleep reported.[41] A similar study of 1845 college students found that students at risk for sleep disorders based on the SLEEP-50 survey were overrepresented among students in academic jeopardy (GPA <2.0).[42]

Limitations

This study implemented the use of a novel validated wearable to measure sleep duration and architecture. Although the device has been well-validated with accuracy that approaches that of polysomnography, polysomnography remains the gold standard for quantification of sleep duration & architecture. In addition, this cohort of sleep data was collected at varying times throughout the year and did not specifically account for athletic competition nor academic schedules. One would anticipate that collegiate athletes' sleep schedules vary in-season versus out-of-season and may vary in the weeks leading up to academic testing.

CONCLUSIONS

Our data confirm that college students and athletes often have poor baseline sleep patterns, with many reporting less than 6 hours of sleep per night. This lack of sleep can have negative consequences on their academic and athletic performance. Further research is needed to understand the factors contributing to poor baseline sleep in these populations and how to improve sleep patterns. These data lay the foundation for optimization of sleep to reduce the risk of medical pathology and enhance school and sports performance.

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